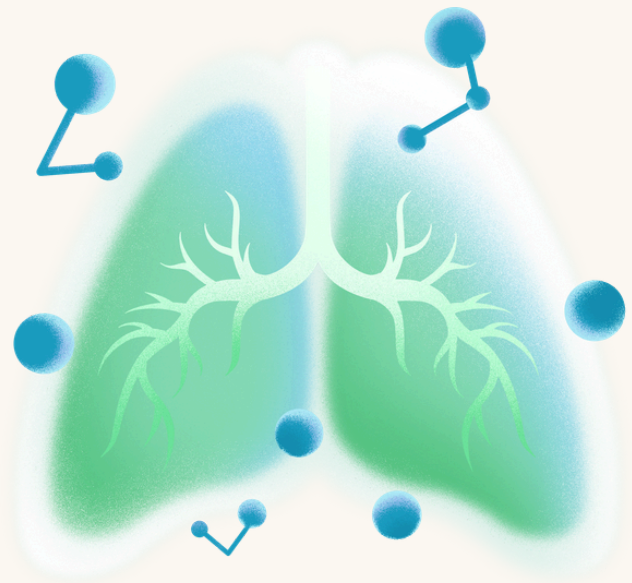


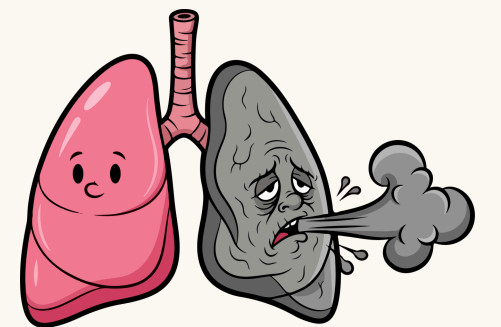


A Data-Driven Approach to

Lung Cancer Risk

Assessment | Classification | Solution

By चस्मा Wizards 



By चस्मा Wizards



DINESH KARKI



SANSKAR DHAKAL



चस्मा Wizards



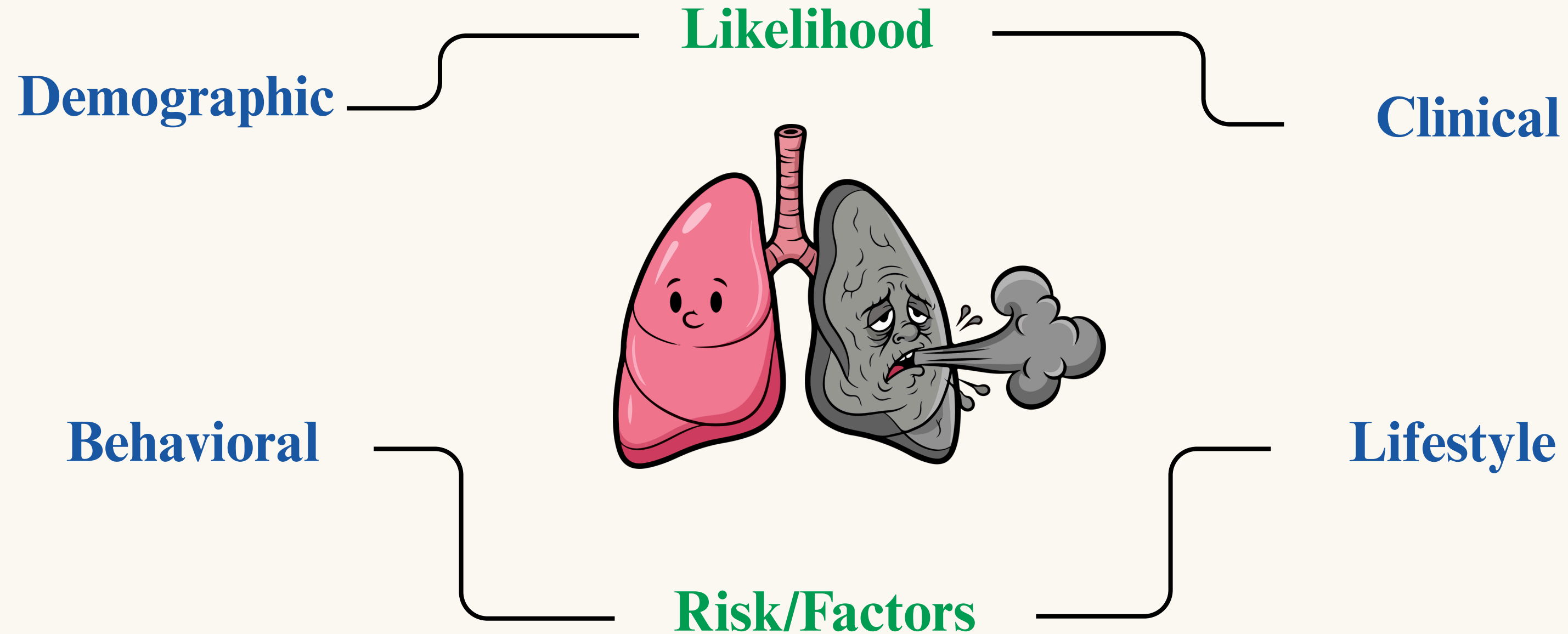
KUSHAL THAPA



MUHAMMAD SHARIF



Problem Statement



Develop a model to predict lung cancer risk using patient data and evaluate how well logistic regression identifies key risk factors.



Importance of the Problem



- *Lung cancer has a high mortality rate, and early detection is difficult; predictive models can help identify high-risk individuals for timely intervention (He, 2023).*
- *Use a regression model to classify lung cancer risk, providing insights into key risk factors through interpretable results.*



Litrature Review: Solution



- *Logistic regression and similar statistical models are reliable, interpretable, and cost-effective tools for lung cancer prediction in real-world scenarios.*
- *Machine learning models, particularly logistic regression, are widely used for accurate lung cancer classification using clinical and laboratory data (Liu et al., 2025).*

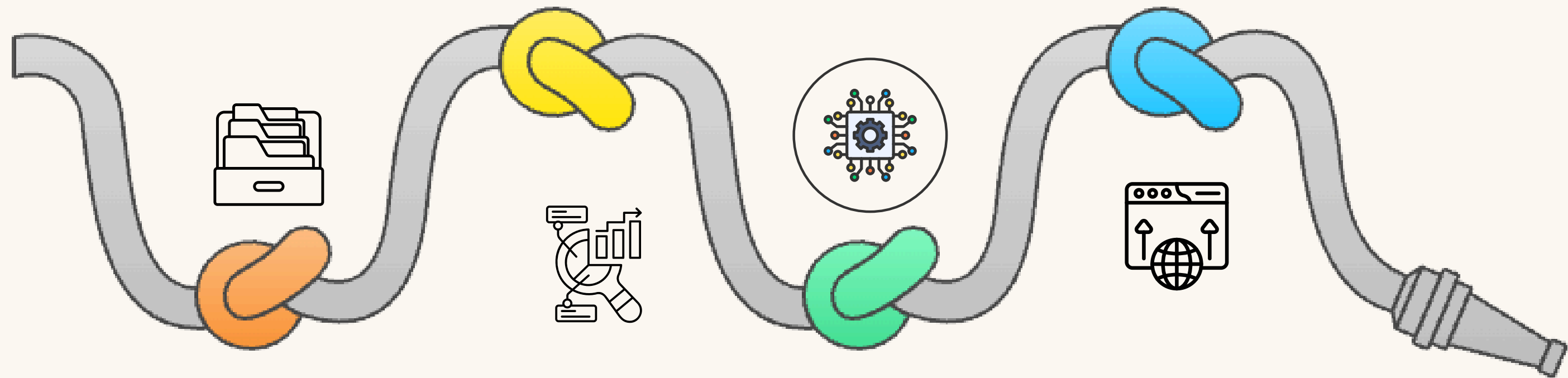
Solution

Data Preparation

Clean, and preprocess the dataset to ensure model readiness.

Model Development

Classify lung cancer risk, and train the model using structured data.



Statistical Exploration

Perform correlation analysis and ANOVA to know the relationships.

Evaluation & Deployment

Evaluate model performance, and develop a simple application.



Data-set



1. Demographic Factors
2. Lifestyle & Environmental Risk Factors
3. Medical History & Symptoms
4. Clinical Measurements



Target Variable

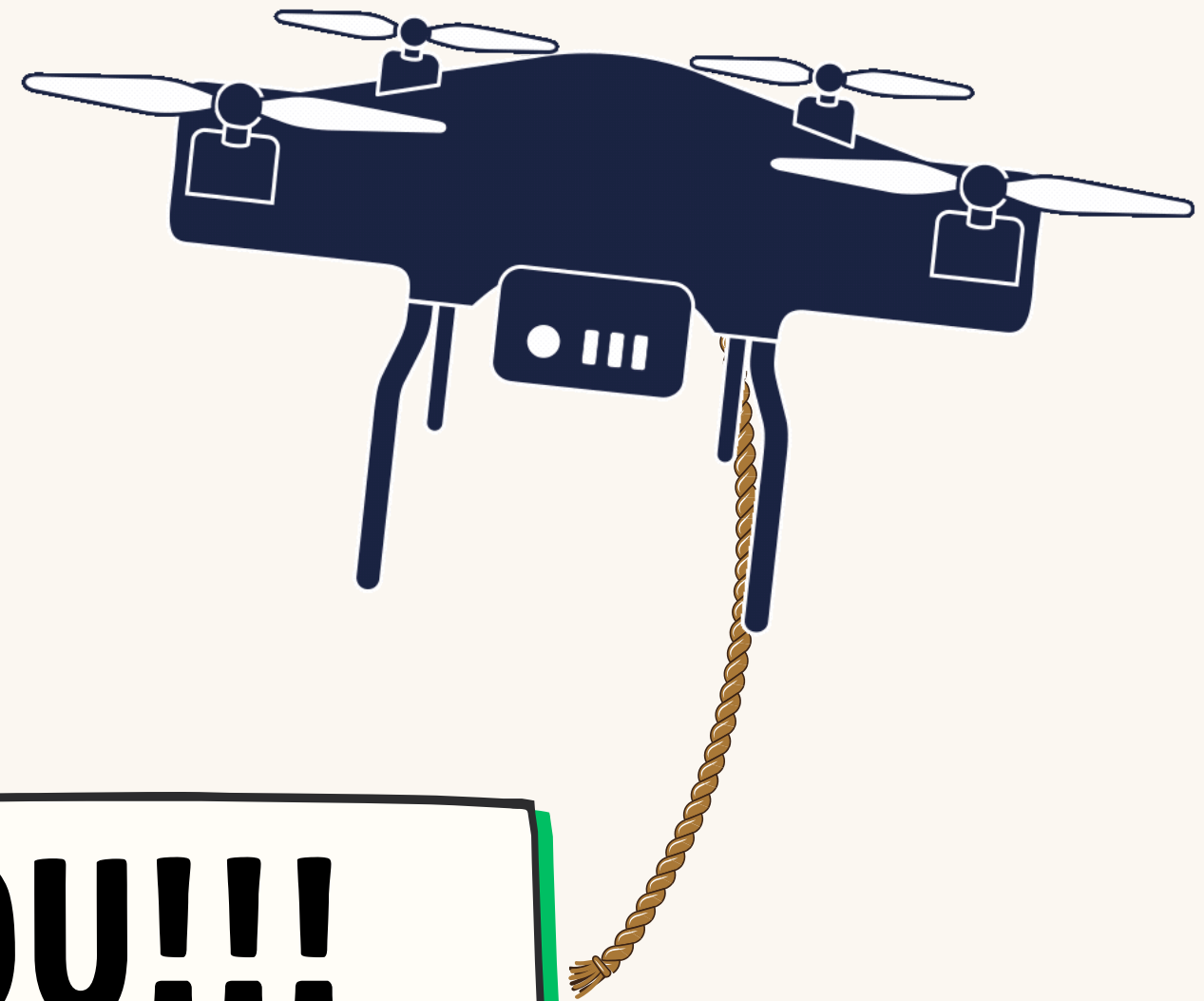
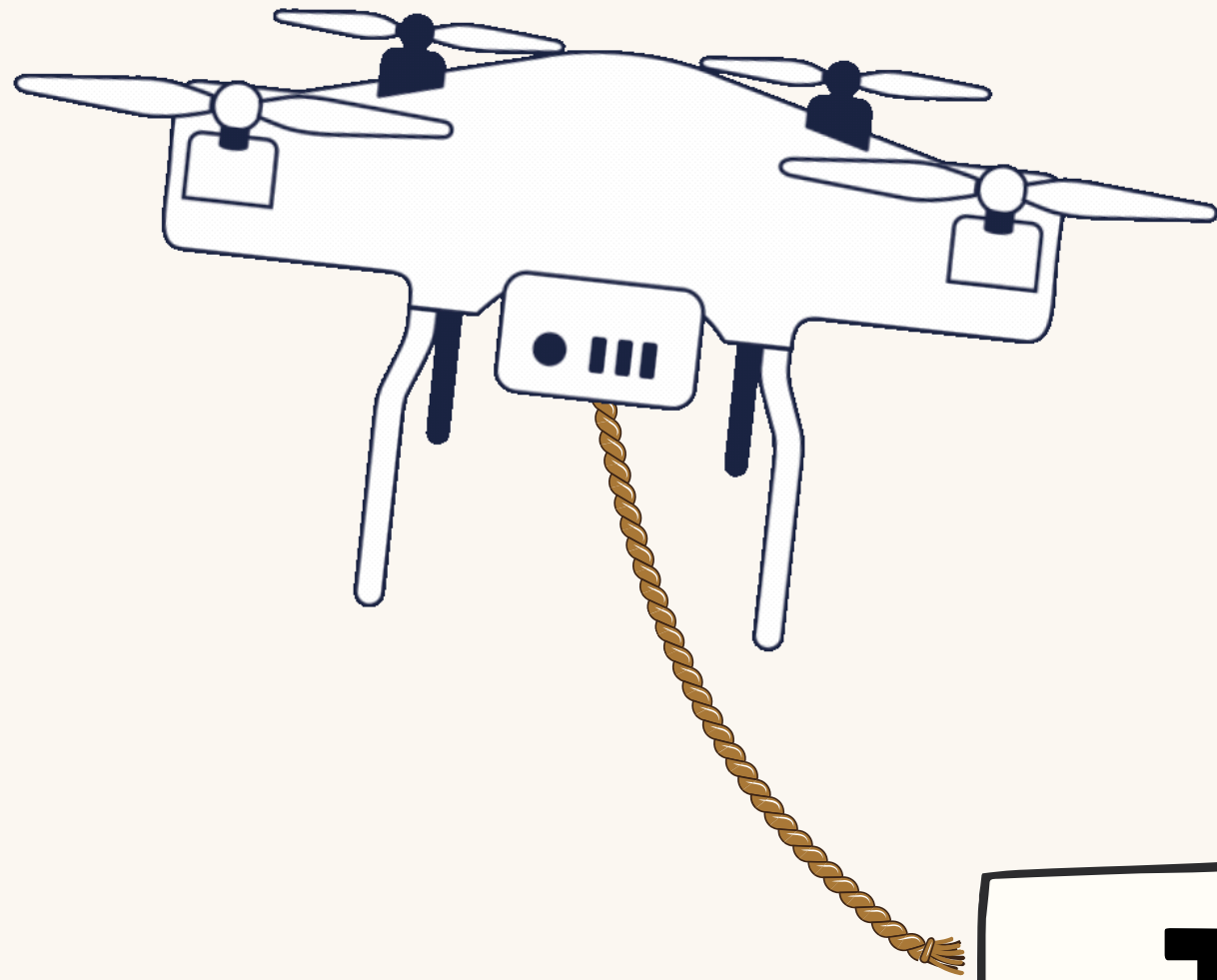
lung_cancer_risk (0/1)



References

Liu, C., Liao, Y., Wang, D., Yang, J., Zhao, L., Liu, X., Wang, Z., & Wu, L. (2025). Machine learning for classification in lung cancer using routine clinical and laboratory data. *Annals of Surgical Oncology*. <https://doi.org/10.1245/s10434-025-18747-y>.

He, W. (2023). Statistical analysis and risk prediction of lung cancer exploiting logistic regression. *Applied and Computational Engineering*. <https://doi.org/10.54254/2755-2721/20/20231105>



THANK YOU!!!

